

Chain rule extra's

$$(inp)^{10} \rightarrow 10(inp)^9 \cdot \frac{df}{dx} \text{ of input}$$

example:

$$(\tan x + 1)^{10} \Rightarrow \frac{dy}{dx} = 10(\tan x + 1)^9 \cdot \frac{df}{dx}$$

$$10(\tan x + 1)^9 \cdot \sec^2 x$$

$$\boxed{\frac{dy}{dx} = 10(\tan x + 1)^9 \cdot \sec^2 x}$$

↑
stays the
same

↑
derivative
of input

$$[\sin(x^3)]^3 \Rightarrow \frac{dy}{dx} = 3 \sin(x^3)^2 \cdot \frac{df}{dx}$$

$$3 \sin(x^3)^2 \cdot 3x^2$$

$$\boxed{\frac{dy}{dx} = 9x^2 \sin(x^3)^2}$$

↑
stays the
same

↑
derivative
of
input

highlighted in yellow is the input

e^{5x} ← input

$$\frac{dy}{dx} = e^{5x} \cdot \frac{dI}{dx} \quad (\text{derivative of the input})$$

$$\frac{dy}{dx} = e^{5x} \cdot 5$$

$$\boxed{\frac{dy}{dx} = 5e^{5x}}$$

explanation:

e^{5x}
↑

keep it the
same doesn't
matter what
the input is

5
↑

the input is
5x so when
derivative the
answer will just
be 5

more examples

→

input

$$y = e^{x^2}$$

keep the same then take input

$$\frac{dy}{dx} = e^{x^2} \cdot \frac{dI}{dx}$$

$$\frac{dy}{dx} = e^{x^2} \cdot 2x$$

$$\boxed{\frac{dy}{dx} = 2x e^{x^2}}$$

input

$$y = e^{\cos x}$$

keep the same then take input

$$\frac{dy}{dx} = e^{\cos x} \cdot \frac{dI}{dx}$$

$$\frac{dy}{dx} = e^{\cos x} \cdot -\sin x$$

$$\boxed{\frac{dy}{dx} = -e^{\cos x} \sin x}$$